# Environment Setup

SQL Server and SSMS for the Data Warehouse project, on Windows and Mac

The guide gets you to the first line of the project: SQL Server installed, the six source files where the scripts look, the database created. With the fixes for the common errors.

## 1 Choose your setup

Pick the path for your computer and read only its pages. Each path starts on its own page and ends with a green box: when you reach it, your setup is done.

| Your computer | Path | Same screens as the videos | Cost | Start |
|---|---|---|---|---|
| Windows | **SQL Server Express + SSMS** | yes | no cost | page 3 |
| Mac | **Docker** and Visual Studio Code | no, but the same SQL | no cost | page 6 |
| Mac | **Parallels** with Windows | yes, SSMS inside Windows | about 100 dollars a year, 14-day trial | page 8 |

### My advice

- **On Windows:** the first path. It is the one in the videos.
- **On a Mac:** Docker. It costs nothing, it is light, and the SQL is the same.
- **On a Mac, with SSMS exactly as in the videos:** Parallels, if your Mac has 16 GB of memory.

### For everyone

- What you get in the materials: page 2
- When something breaks: page 9

## 2 What you get in the materials

Download the project materials once. You get this folder:

```
sql-data-warehouse-project/
├── 00_setup/                this guide, and the one-click Docker start for Mac
├── 01_datasets/             the six source files the project loads
│   ├── source_crm/          three files from the CRM system
│   └── source_erp/          three files from the ERP system
├── 02_docs/                 the requirements, the diagrams and the notes
├── 03_scripts/              the project scripts, in the order of the videos
│   ├── init_database.sql    creates the database: run it first
│   ├── bronze/              loads the raw files
│   ├── silver/              cleans and prepares the data
│   └── gold/                builds the tables for analysis
└── 04_tests/                checks the quality of the silver and gold data
```

The project builds one database, `DataWarehouse`, with the schemas `bronze`, `silver` and `gold`. The scripts are the ones written in the videos, so you can type along or run them.

> The videos show the folders without numbers. `datasets` there is `01_datasets` here, `scripts` is `03_scripts`, and the scripts already use the new names.

# Windows

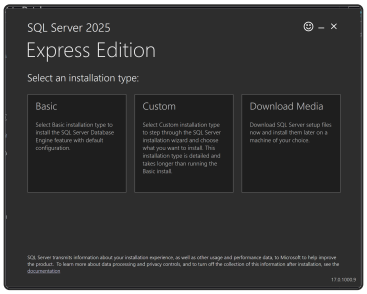
## 3 Install SQL Server and SSMS

SQL Server Express costs nothing, and it is all you need. The videos show SQL Server 2022. The download page now offers 2025, and it works the same.

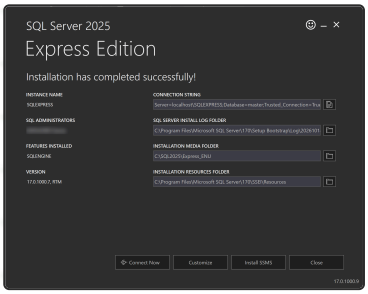
- Download SQL Server Express:
  - Open the SQL Server downloads page.
  - Scroll to the **Express** card.
  - Click **Download now**.
- Run the downloaded file:
  - Choose **Basic**.
  - Accept the licence and click **Install**.
- The last screen shows your server name: `localhost\SQLEXPRESS` . Keep it, you need it in a minute. Your Windows account is now the administrator of this SQL Server.
- Download SQL Server Management Studio (SSMS) and install it:
  - Keep the default options.
  - The installer may offer extra components. You need none of them.




Step 1: the Express card.




Step 2: choose Basic.

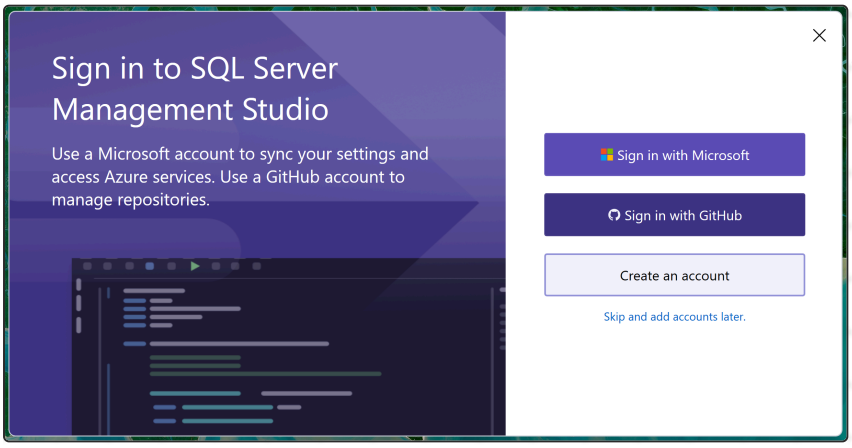



Step 3: the server name.

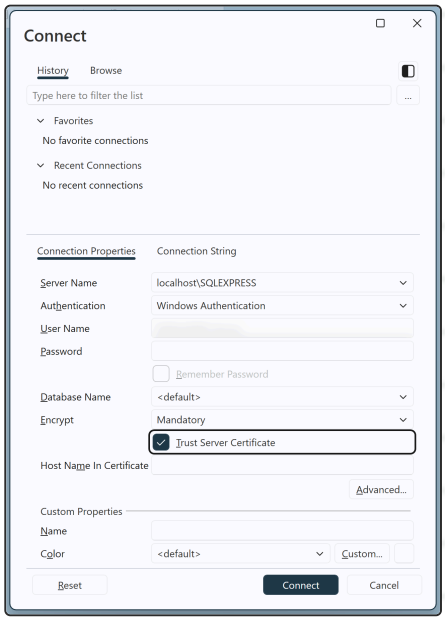
## Connect for the first time

SSMS 21 and newer look different from the videos.

- At the first start, SSMS asks you to sign in. You do not need an account: click **Skip and add accounts later** under the buttons.
- In the **Connect** window, fill in three fields:
  - **Server name:** `localhost\SQLEXPRESS` . If the field is empty, that is normal.
  - **Authentication:** Windows Authentication.
  - **Trust Server Certificate:** ticked. Without it you get a certificate error.
- Click **Connect**. Your server appears on the left, in the Object Explorer.




Step 1: the skip link under the buttons.




Step 2: the Connect window, filled in.



DATA WITH BARAA

## 4 The files and the database

Two things before the first lesson of the project: the six files where the loader looks for them, and the empty database.

### Put the files where the scripts look

- Create the folder `C:\sql\dwh_project`.
- Copy the `01_datasets` folder into it. The six files now sit where `proc_load_bronze.sql` reads them:
  - `C:\sql\dwh_project\01_datasets\source_crm\`: cust_info.csv, prd_info.csv, sales_details.csv
  - `C:\sql\dwh_project\01_datasets\source_erp\`: CUST_AZ12.csv, LOC_A101.csv, PX_CAT_G1V2.csv
  - **Typing the loader while you watch?** Write `01_datasets` where the video says `datasets`.
- Keep the folder where SQL Server can read it:
  - **Works:** a folder directly on `C:\`.
  - **Often fails:** Downloads, Desktop, OneDrive. The load then stops with "Access is denied".
  - **Somewhere else:** change the six paths in `proc_load_bronze.sql` to your folder.

### Create the database

- In SSMS, open `03_scripts/init_database.sql`:
  - Click **File > Open > File** and pick the script.
  - Click **Execute**, or press F5.
  - Refresh the Databases folder: `DataWarehouse` is there, with the schemas `bronze`, `silver` and `gold`.
- For every script after this one, pick `DataWarehouse` in the database dropdown of the toolbar, where it says `master`. Otherwise the script stops with "schema name bronze does not exist".
- From here, follow the project:
  - The bronze lesson creates the tables and the loader.
  - `EXEC bronze.load_bronze` fills them from the six files.

The script drops and recreates `DataWarehouse` every time it runs. Run it once, at the start. If you run it again later, your bronze, silver and gold work is gone.

**Done.** The six files sit in `C:\sql\dwh_project\01_datasets`, and `DataWarehouse` is ready for the bronze lesson.

# Mac with Docker

## 5 Start SQL Server in Docker

SQL Server does not install on macOS, and SSMS is Windows only. With Docker, SQL Server runs in a small container on your Mac, and you write your queries in Visual Studio Code. The SQL is the same, BULK INSERT and stored procedures included.

### Install, once

- Install Docker Desktop, the Apple chip version. Start it and leave it running.
- Open the Docker Desktop settings, **General**. SQL Server ships for Intel chips only, and these two settings let your Mac run it:
  - **Virtual Machine Manager:** Apple Virtualization framework.
  - **Use Rosetta for x86_64/amd64 emulation on Apple Silicon:** on.
- Install the tool you write your queries in:
  - Visual Studio Code.
  - Inside it, the extension SQL Server (mssql) by Microsoft. It is Microsoft's tool for the Mac.

### Way 1: one click

- In the project materials, open `00_setup/mac-docker` and double-click `start.command`.
  - **macOS blocks it?** Open System Settings > Privacy & Security, scroll down, click **Open Anyway**, then double-click again. Only the first time.
- Wait a minute. The first start downloads SQL Server. Then the command copies the six files into SQL Server, to the folder the loader reads.
- Nothing to edit: the scripts run exactly as in the videos.
- Next time, double-click the same file again.

OR

## Way 2: by hand

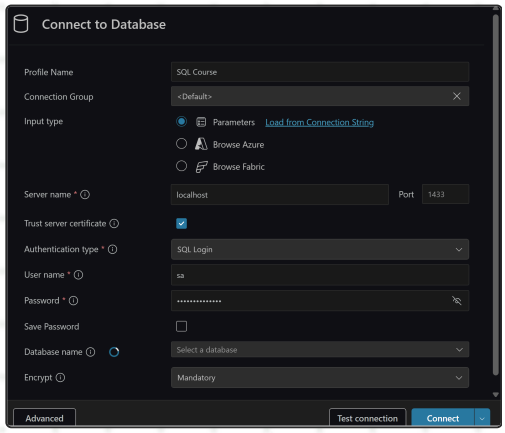
Start SQL Server, then copy the datasets to the place the loader reads. Choose your own password: at least 8 characters with upper case, lower case, a number and a symbol.

```
docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=YourStrong!Passw0rd" \
  -p 1433:1433 --name sql --platform linux/amd64 \
  -d mcr.microsoft.com/mssql/server:2022-latest
docker exec -u root sql mkdir -p /sql/dwh_project
docker cp path/to/sql-data-warehouse-project/01_datasets sql:/sql/dwh_project/
docker exec -u root sql chown -R mssql /sql
```

Microsoft's own guide for the first command: SQL Server in a Docker container.

## Connect and create the database

- In Visual Studio Code, open the SQL Server extension on the left and add a connection:
  - **Server name:** `localhost`
  - **Port:** `1433`
  - **Trust server certificate:** ticked
  - **Authentication type:** SQL Login
  - **User name:** `sa`
  - **Password:** `SqlServer!2026`, or your own from way 2
- Open `03_scripts/init_database.sql` and run it. It creates `DataWarehouse`.
- For every script after it, switch to `DataWarehouse`: click the database icon at the top right of the editor and pick it.




The connection form, filled in.

**Done.** The six files sit where the loader reads them, and `DataWarehouse` is ready for the bronze lesson.

# Mac with Parallels

## 6 Install Windows in Parallels

This path gives you SSMS exactly as in the videos, because you run Windows on your Mac.

- Install Parallels Desktop. It downloads and installs Windows 11 for you in a few clicks.
  - **Parallels:** about 100 dollars a year, with a 14-day trial.
  - **Windows:** needs its own licence.
- Give the virtual machine memory:
  - **At least:** 6 GB for Windows.
  - **Your Mac:** 16 GB is the comfortable minimum.
- Start Windows in Parallels. From here on, you set up SQL Server like any Windows user.

SQL Server Express installs and runs on the ARM version of Windows 11, even though Microsoft does not list it as supported. Tested with SQL Server 2025 Express and SSMS 22 in Parallels on an Apple chip.

**Next:** inside Windows, follow the Windows path from page 3.

## 7 When something breaks

Find your message in the first column. The fix is in the last one.

### Install and connect

| What you see | Why | What to do |
|---|---|---|
| SSMS asks you to sign in at the first start | optional account | Click **Skip and add accounts later** |
| The server name field is empty | normal | Type `localhost\SQLEXPRESS` |
| Error 26, "Error Locating Server/Instance Specified" | SQL Server is not running, or the name is wrong | Open **Services**, start "SQL Server (SQLEXPRESS)". If you installed the Developer edition, the server name is just `localhost` |
| "The certificate chain was issued by an authority that is not trusted" | SSMS encrypts the connection | Tick **Trust Server Certificate** |
| Installer: "Wait on the Database Engine recovery handle failed" | SQL Server could not start during setup | Uninstall SQL Server, restart Windows, install again. If it fails again, the real reason is in the ERRORLOG file under `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Log` |
| SQL Server 2025 setup stops at the rules check | installer problem on some machines | Install SQL Server 2022 Express instead. Everything here runs on it |
| Your laptop does not let you install anything | no administrator rights | SQL Server needs them. Ask IT, or use a personal computer |

### The database and the files

| What you see | Why | What to do |
|---|---|---|
| Msg 262, "CREATE DATABASE permission denied in database 'master'" | your Windows login is not an administrator of this SQL Server. Someone else installed it, often IT | Log in with the Windows account that installed SQL Server, or ask an administrator to add your login to the **sysadmin** role (Security > Logins > Properties > Server Roles) |
| Msg 2760, "The specified schema name 'bronze' either does not exist" | the query runs in `master`, not in `DataWarehouse` | Switch the database to `DataWarehouse`, then run the script again. If the database is missing, run `init_database.sql` first |
| Msg 4860, "Cannot bulk load. The file does not exist or you don't have file access rights" | the file is not at the path in the script | The loader reads `C:\sql\dwh_project\01_datasets\...`, not `datasets` as in the videos. Put the folder there, section 4, or change the six paths in `proc_load_bronze.sql` |
| Msg 4861, "Cannot bulk load because the file could not be opened. Operating system error code 5 (Access is denied.)" | SQL Server runs as a service and cannot read your folder | Move the folder to `C:\sql\dwh_project\`. Downloads, Desktop and OneDrive folders are the usual culprits |
| Msg 208, "Invalid object name 'bronze.crm_cust_info'" | the bronze tables are not created yet | Run `ddl_bronze.sql` in `DataWarehouse`, then the loader |

| What you see | Why | What to do |
|---|---|---|
| Msg 4864 or 4832, a conversion error or an unexpected end of file during the bulk load | a CSV file was changed, usually by Excel saving it | Copy the original file from the package again. Open CSV files with a text editor, not with Excel |
| A .sql file opens in Notepad | the file is not linked to SSMS | Open SSMS first, then **File > Open > File** |

### Docker

| What you see | Why | What to do |
|---|---|---|
| macOS: "start.command" cannot be opened, or "Apple could not verify" it | the file comes from the internet and is not signed | System Settings > Privacy & Security, click **Open Anyway**, then double-click again |
| "Cannot connect to the Docker daemon" | Docker Desktop is not running | Start Docker Desktop, wait for the whale icon, try again |
| "no matching manifest for linux/arm64" | SQL Server is built for Intel chips | Turn on Rosetta in Docker Desktop, section 5. By hand, add `--platform linux/amd64` to the command |
| "port is already allocated" | another SQL Server listens on 1433 | Stop it, or change the first `1433` in `docker-compose.yml` to `1434` and connect to port `1434` |
| The container stops right after the start | the password breaks the rules | At least 8 characters with upper case, lower case, a number and a symbol |
| "SQL Server did not answer" from start.command | SQL Server could not start, most often Rosetta is off | Docker Desktop settings, **General**, turn on Rosetta, section 5. The container's log in Docker Desktop names the reason |

If you are still stuck, write in the Q&A with the exact error message and a screenshot of the window. Then I can see what happened.